

4-Bit Signed ALU - Project Overview

This document gives the short project-level overview of the completed 4-bit signed ALU. The ALU was designed in simulation and then implemented using logic ICs on breadboards, with an Arduino readout used only as a display interface.

Project Goal

The goal was to build a working 4-bit signed two's complement ALU using real digital logic ICs. The ALU supports addition, subtraction, signed comparison, and signed multiplication. It accepts two 4-bit inputs, A and B, and produces operation-specific outputs.

Signed Number Format

Inputs are 4-bit two's complement values. The valid signed range is from 1000 to 0111, which represents -8 to +7.

```
A[3:0], B[3:0]
1000 = -8
1111 = -1
0000 = 0
0111 = +7
```

Main Operations

Operation	Control / Block	Output
Addition	SUB = 0 in ADD/SUB block	4-bit sum, Cout, overflow flag
Subtraction	SUB = 1 in ADD/SUB block	4-bit difference, Cout, overflow flag
Comparison	74HC85 unsigned comparator + sign correction	GT, EQ, LT
Multiplication	absolute value -> unsigned 4x4 multiplier -> signed 8-bit product	8-bit signed product

Arduino Display

The Arduino Nano is used as an output display/readout layer. It reads the ALU output lines through 74HC165 parallel-in serial-out shift registers and shows the selected result on an LCD1602 I2C display. It does not calculate the ALU result; the logic ICs perform the computation.

Build Summary

The physical build uses 74HC/74LS logic ICs, DIP switches for inputs and controls, LEDs for bit-level debugging, decoupling capacitors for stable power, and an Arduino/LCD display for a cleaner final readout. Each major block was tested separately before being connected into the full ALU.